Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Recommendation Scores

> Prepare Recommendation Scores: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Compute a **weight**ed **score**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and **rank training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> **Score**=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

Concept to explain

A **weight**ed **score** expresses priorities but also embeds choices.

Term	Meaning
weight	A multiplier controlling how strongly a feature or value contributes to a calculation.
score	A numerical value assigned using a specified rule; it is not automatically a probability.
rank	Place candidates in an order using a declared comparison rule.
tradeoff	A situation where improving one desired property can make another harder to achieve.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 23 as a two-session investigation using fictional data. A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning.

The activity must also take at most 20 minutes. Duration is a hard requirement; a high **score** cannot compensate for exceeding it. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

A gap of 2.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can **rank** only eligible options with a transparent **weight**ed **score**.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Recommendation Scores using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Score** = 3×genre match + 2×difficulty match + 1×length match.

2. Book A matches genre and length: 3+0+1=4.

3. Book B matches difficulty and length: 0+2+1=3.

4. A **rank**s first for these **weight**s—not as a universal best book.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Score=2 times genre match + length match. A matches both, B genre only, C length only.

The teaching case

Score and **rank**.

Exact explanation to model

Why the result follows

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Explain the added worked example

A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning. The activity must also take at most 20 minutes.

Duration is a hard requirement; a high **score** cannot compensate for exceeding it.

Only A is printed, with **score** 10. If B took 20 minutes it would qualify and have the higher **score** 11. The inclusive boundary matters.

Ratings and **weight**s are invented for practice and require justification before use in a real decision.

Key terms

Term	Meaning
Weight	A multiplier expressing the importance assigned to a feature
Score	A number calculated to compare options
Constraint	A requirement an option must meet
Tradeoff	Improving one preference while giving up another

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **If enjoyment weight becomes 1 and learning weight becomes 2, find both scores before applying the same duration limit.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Recommendation Scores

A fictional book-activity chooser has a 14-minute limit.

The facts we will use

> **Supplied facts**
>
> **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

> **Supplied facts**
>
> Require whole nonnegative feature values; time must be at most the limit. Highest **score**

wins; tie by ID.

AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/23

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A:2 x 3+2=8. C:2 x 2+5=9.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Eligibility first	A and C fit time; B takes 15 and is excluded.
2. Scores	A:2×3+2=8. C:2×2+5=9.
3. Choose	C wins at score 9 among eligible activities.
4. Explain exclusion	B's score 11 cannot override its failed time requirement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

What this example does not establish

Keep the stated limit

Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Change one thing in the pictured case

Change one thing

Change: Change interest **weight** from 2 to 3, keeping time limit 14.

Predicted result

Prediction: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Why the result follows

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Ask while showing the picture

Ask while showing the picture · 1

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Recommendation Scores, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Score**=2 times genre match + length match. A matches both, B genre only, C length only.

2. State the decision to explain

Write the question to explain: **Score** and **rank**.

3. Reveal the worked reasoning

Reveal after **prediction**s: A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Hold back the independent task

Keep for an independent attempt: A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

A **score**s 8 and B **score**s 13. B remains excluded at 25 minutes. Exit: First **score**s A 9, B 8 so A wins. Second **score**s A 6, B 10 so B wins. Explain the preference change.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.

Stage	Teacher action
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Which durations 19, 20, 21 qualify under at most 20?

Reasoning to listen for

Teacher check: 19 and 20 qualify; 21 does not.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why does B not become the winner when its **score** is largest?

Reasoned teacher answer

Its time 15 exceeds the 14-minute maximum, so it is ineligible before **rank**ing.

Changed-case reasoning: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Recommendation Scores investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Create two reader profiles with different **weight**s, calculate every term, total, **rank**, and explain why the order changes.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation.

Avoid claiming that a larger **score** proves greater educational value.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: limit 10, X (fun 1, learning 5, minutes 9), Y (3,2,10), Z (5,5,11). **Score**s 7, 8, 15; choose Y, exclude Z.

A convenient shortcut preserves meaning: Choosing the highest **score** and only then ignoring its duration breaks the requirement. Filter for eligibility first.

If no item qualifies, report no eligible option instead of relaxing a requirement silently.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Recommendation Scores questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Independent question

Independent question · withhold the key

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

Expected independent answer

Teacher key · Expected independent answer

A **score**s 3 and B **score**s 1, so A **rank**s first. The **rank**ing reflects the chosen **weight**s; different priorities could change them.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Recommendation **score**s are objective preferences of all readers.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Scores A=4, B=3, C=5. With illustrated required, C is excluded and A wins. A high **score** cannot compensate for a mandatory requirement unless the requirement has explicitly changed.

Explanations for the additional workbook examples

W1 Calculate a score

Workbook question · W1

An eligible activity has fun 2 and learning 4. Use 2 * fun + learning.

Teacher reasoning · W1

Score 8.

Teacher reasoning · W1

Substitute each rating into its named role: 2 * fun + learning = 2 * 2 + 4 = 4 + 4 = 8. Eligibility is supplied in this task, so this calculation **rank**s an already permitted option.

The **score** is a result of the chosen preference rule, not a measured educational benefit.

W2 Check a boundary

Workbook question · W2

Which durations 19, 20, 21 qualify under at most 20?

Teacher reasoning · W2

19 and 20 qualify; 21 does not.

Teacher reasoning · W2

At most 20 means minutes <= 20. The value 19 is below the limit and 20 is exactly at it, so both qualify. The value 21 is above it and is excluded.

An answer that rejects 20 has changed the inclusive requirement into a strict one.

W3 Repair selection

Workbook question · W3

An ineligible item **score**s 15 and an eligible item **score**s 9. Which should be offered under the stated policy?

Teacher reasoning · W3

Offer the eligible item with **score** 9. The **constraint** is applied before **rank**ing.

W4 Choose independently

Workbook question · W4

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

Teacher reasoning · W4

Scores A 10, B 11, C 9. B is excluded. A is highest among eligible options.

W5 Test sensitivity

Workbook question · W5

For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare **weight**s (2,1) with (1,2).

Teacher reasoning · W5

First **score**s A 9, B 8 so A wins. Second **score**s A 6, B 10 so B wins. Explain the preference

change.

Assess independent transfer

Independent question · withhold the key

Selected case: Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

Selected case answer

Teacher key · Selected case answer

Scores A 10, B 11, C 9. B is excluded. A is highest among eligible options.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Recommendation Scores to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Publish the features, **weight**s, raw **score**s, and reason for the top result.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 24: Feedback Loops. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use separate eligibility and **score** columns. Extension: add a documented tie rule and test tied eligible items.

Independent W4 **evidence**: **Score**s A 10, B 11, C 9. B is excluded. A is highest among eligible options. Record whether the learner needed no help, a hint or explanation.

Fresh retry: limit 10, X (fun 1, learning 5, minutes 9), Y (3,2,10), Z (5,5,11). **Score**s 7, 8, 15; choose Y, exclude Z.

Next week: Implementing a recommender. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/24, ai-2/26, ai-2/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Recommendation Scores lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Compute a **weight**ed **score**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Which durations 19, 20, 21 qualify under at most 20?

Supported practice answer

Reasoned practice answer: 19 and 20 qualify; 21 does not.

Check every learner before discussion: Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Weight**s represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Change interest **weight** from 2 to 3, keeping time limit 14. Require a **prediction** and a reason before comparing. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

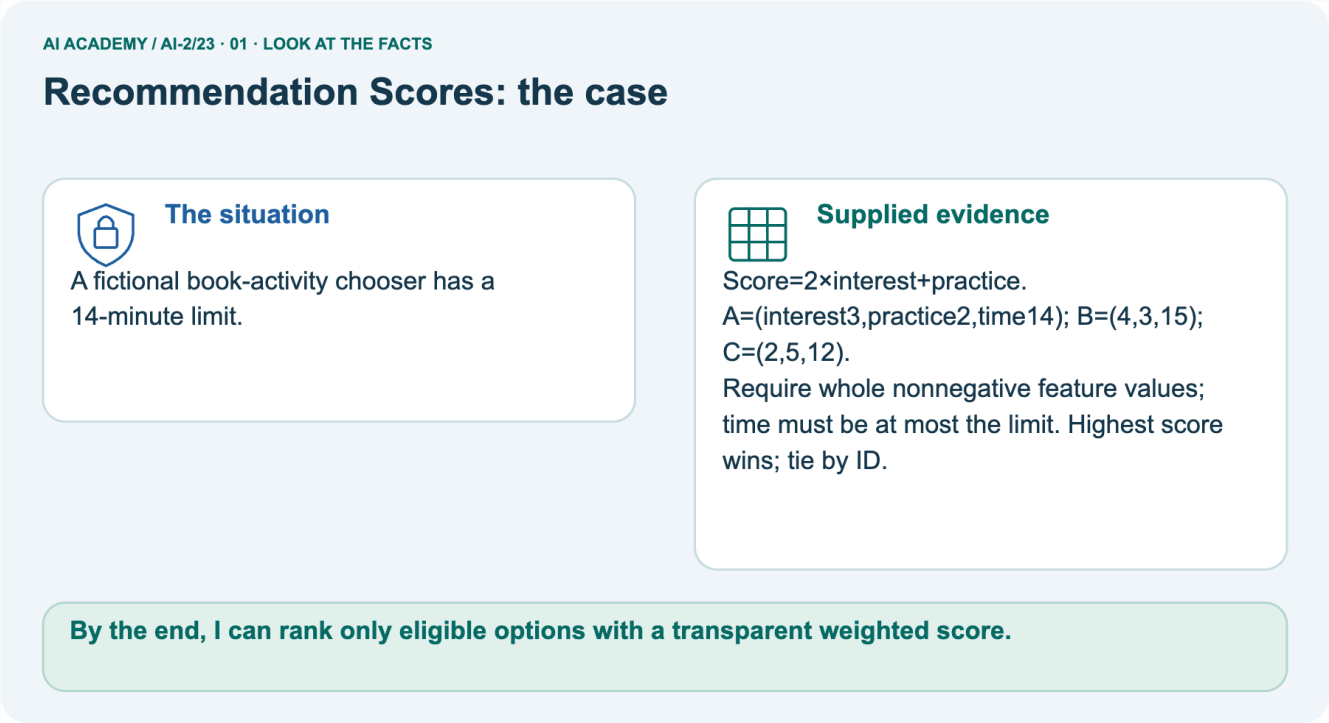
TEACHER GUIDE · 9 OF 9

Visual concept journey

> Show one Recommendation Scores graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Recommendation Scores: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

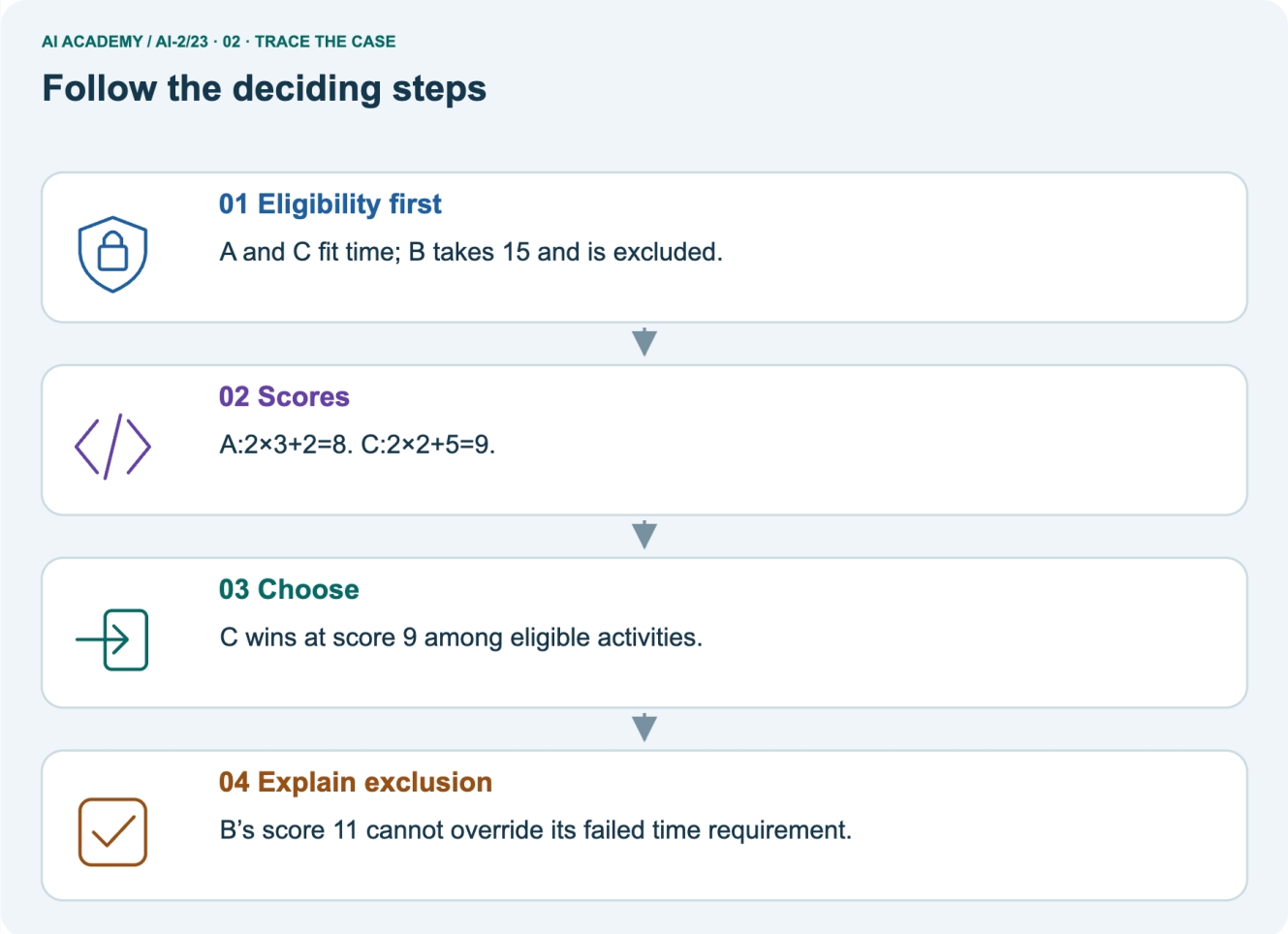
Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Time filtering and scores



Follow the labelled picture with the matching explanation. Open the vector diagram

Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

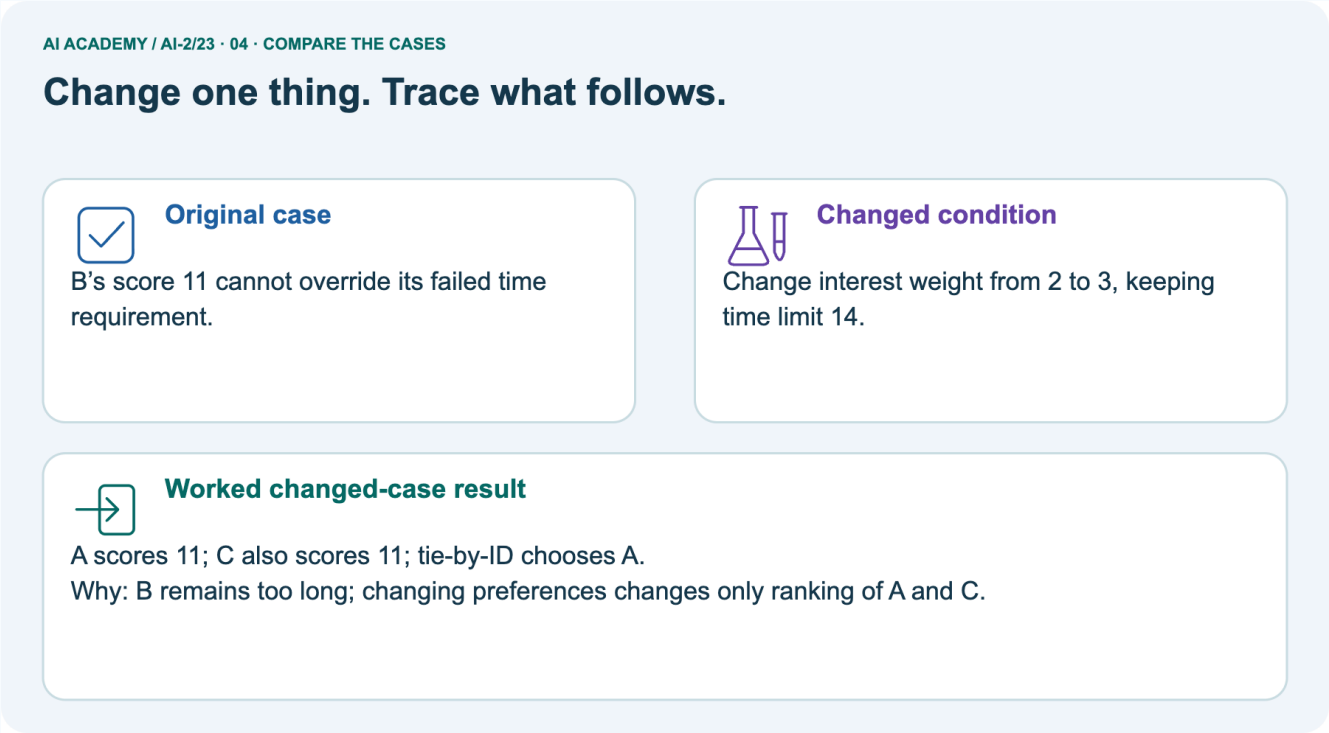
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

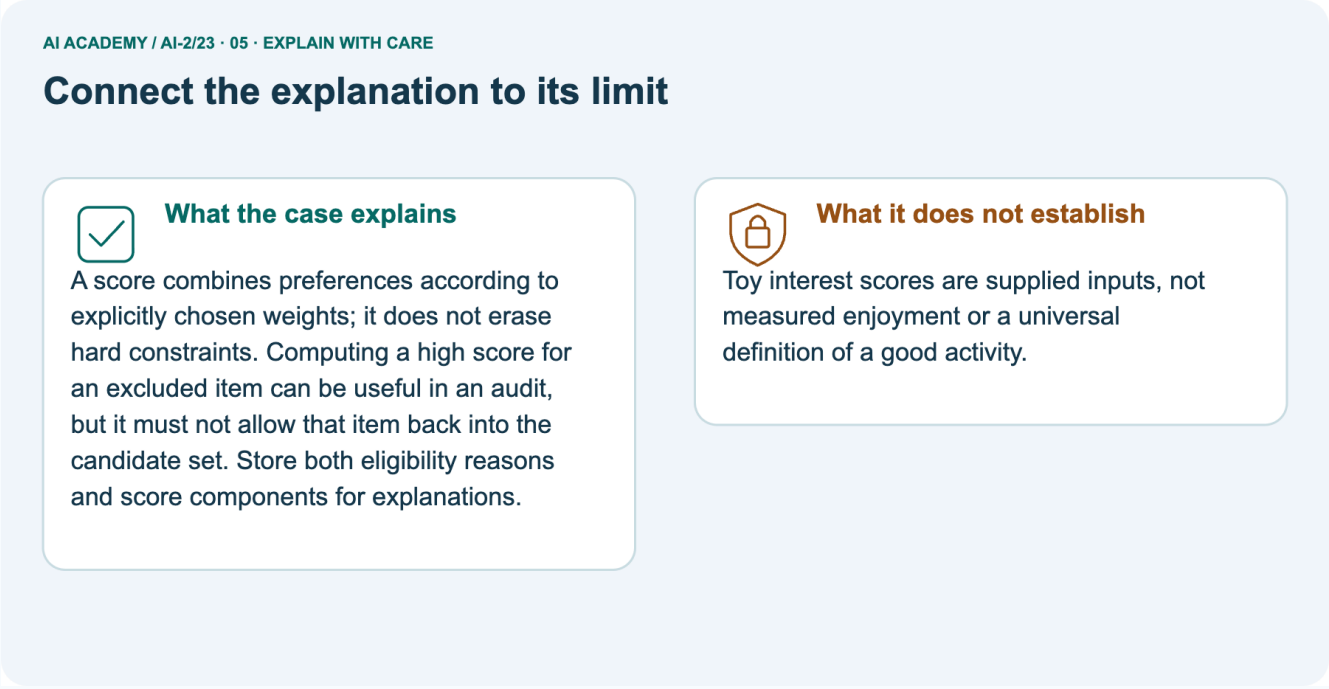
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: B remains too long; changing preferences changes only **rank**ing of A and C.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

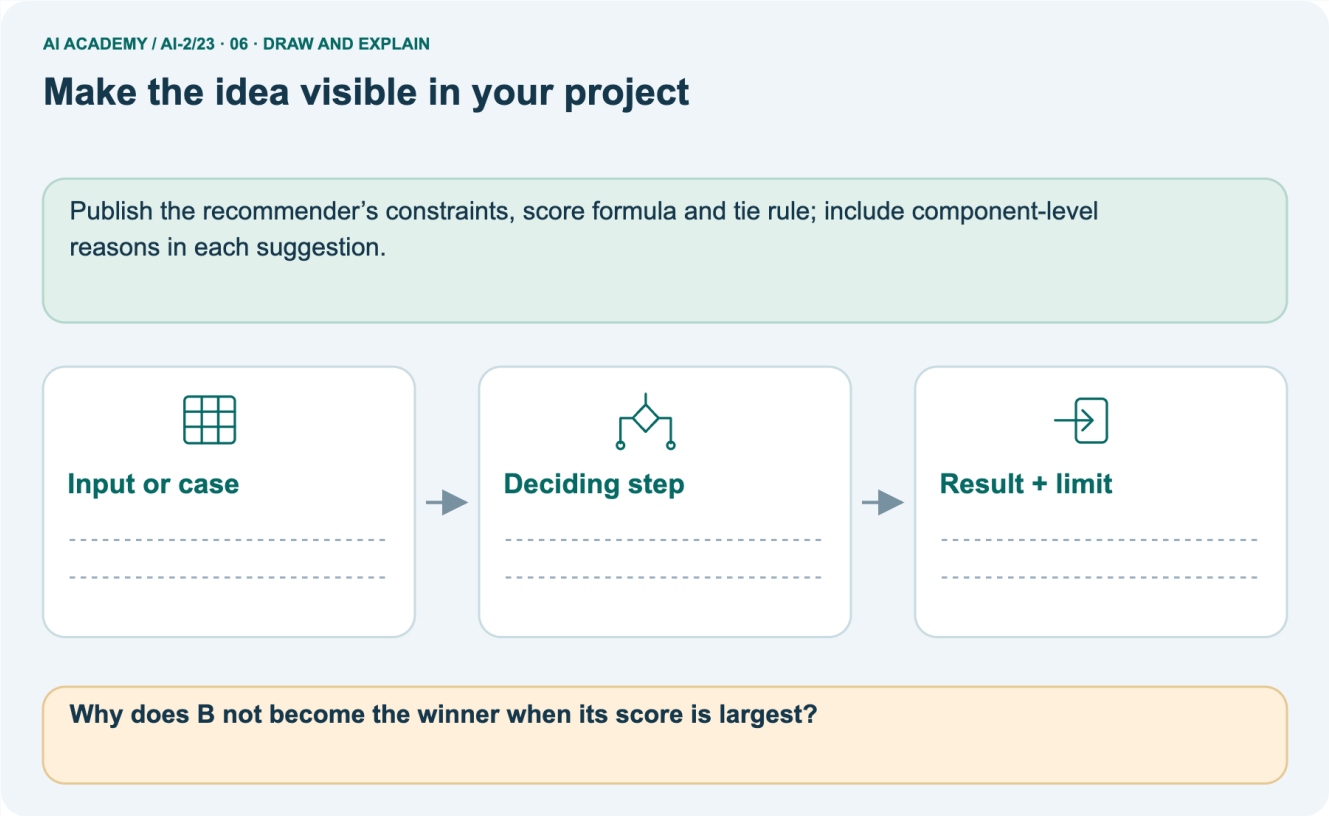
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Its time 15 exceeds the 14-minute maximum, so it is ineligible before **rank**ing.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.